

Agricultural Modular Courses: Agro-Ecology and Community Development

At one week, two weeks, one month, three months, nine months, one year modular courses

1. Animal production

- **Introduction to animal production**
 - Covers the importance of livestock in food security and economy
 - Animal behaviour
 - Different animal production systems (Intensive, Extensive, and organic systems).
 - Overview of livestock types
 - Importance of livestock industry
 - Differences between indigenous and exotic breeds.
- **Animal Nutrition and Feeding**
 - Studies digestive systems (Digestive physiology of ruminants vs. Non – ruminants).
 - Nutrient requirements (carbohydrates, proteins, minerals, vitamins) for growth, reproduction, and production
 - Feed types (forages, concentrates and supplements)
 - Feed formulation and ration formulation techniques for different livestock species.
 - Fodder conservation (hay/silage)
- **Breeding and Genetic**
 - Introduction to Mendelian laws
 - Genetic principles: heredity, genotype, and phenotype
 - Selection methods and breeding systems (purebred, cross breeding).
 - Reproductive technologies: Artificial insemination (AI) and oestrus synchronization, and improving economic traits.
 - Record keeping and performance recording
- **Livestock Management Practices**
 - Routine management: Feeding, watering, handling, and grooming
 - Routine duties: Hoof trimming, docking, dehorning, castration, shearing, identification methods, and milking procedures.
- **Animal Health and Disease Control**
 - Preventive health care measures, vaccines, parasite control (dipping/spraying), and bio security measures
 - Zoonoses and “One Health” concept
 - Common diseases.
- **Production Systems (species – specific)**

- **Housing and Environment:** Design and management of animal shelters, ventilation, hygiene, and environmental impact on production.
- **Dairy Production:** Udder structure, milking procedures, and milk handling.
- **Poultry Production:** Fowl/poultry breeds, broiler, and layer management.
- **Beef and small Ruminants:** Sheep, goat, and camel management.
- **Pig Production:** Housing, feeding, and breeding.
- **Analysis of Production Systems:** Intensive vs. extensive systems, and organic animal production.
- **Agribusiness and Technology**
 - Economics and Marketing: Record keeping, business planning, farm finance,
 - Farm mechanization and structures.
 - Marketing and value addition of animal products.
 - Use of farm management software and computer applications.
 - Environmental impact and sustainability.

Key Skills Developed

- Gain practical skills in animal handling, vaccination, feed formulation, and farm business management.

Practical and Field Components

- Farm machinery maintenance and housing construction.
- Identification of farm machinery and tools
- Weight determination and livestock handling.
- Field attachment/internships (Industry experience)
- Practical feed formulation using local resources

Expected Outcomes

- This type of course is often structured into comprehensive units for certificates or diplomas in agriculture and animal science.
- Graduates will be prepared to manage livestock enterprises, improve animal productivity through scientific methods, and work in agricultural extension services, aiming for sustainable development and self-employment.

2. Environmental Science

An Environmental science course provides an interdisciplinary foundation in ecology, biology, chemistry, and policy, focusing on sustainability, biodiversity, pollution management, and natural resource conservation. Core topics include ecosystem structure, climate change, energy systems, and environmental regulations, designed to equip students with analytical skills to solve complex environmental challenges.

Typical Course Curriculum

- **Introduction to Environmental Science**
 - Fundamental principles
 - Environmental history
 - Earth systems (atmosphere, biosphere, hydrosphere, geosphere), and sustainability
- **Ecosystems and Biodiversity**
 - Structure and function
 - Energy flow
 - Food webs
 - Ecological succession
 - Conservation biology
- **Natural Resource Management**
 - Renewable and non-renewable resources
 - Forest management
 - Water resources
 - Mineral extraction, and
 - Agriculture
- **Pollution and Environmental Health**
 - Types of pollution (water, air and soil)
 - Waste management
 - Toxic contaminants, and
 - Environmental impact assessments.
- **Climate change and Policy**
 - Global Climate feedbacks
 - Renewable energy sources
 - Sustainable development, and
 - Environmental law
- **Research Methods and Technology**
 - Laboratory techniques
 - Data analysis
 - Geographic Information Systems (GIS), and
 - Remote sensing

Key Learning Outcomes

- Analyze ecological processes and human impacts on the environment
- Evaluate sustainability strategies at local, national, and global levels
- Apply scientific methods and laboratory techniques to environmental research

- Develop skills in environmental policy analysis and management

Typical Course curriculum

- **Duration:** At one week, two weeks, one month, three months, nine months, one year modular courses
- **Assessment:** Mix of lectures, laboratory experiments, field studies, research projects, and examinations
- **Requirements:** Background in biology, chemistry, mathematics, or geography.

3. Agro-ecology and Agro-biodiversity Course

An Agro-ecology course typically covers sustainable agricultural practices, soil health, biodiversity, and social-ecological systems to create resilient food systems. Key topics include ecological principles, soil management, pest control, agroforestry, and socioeconomic aspects like food sovereignty and market access, suitable for interns, students, farmers, and development practitioners.

Typical Course Curriculum

- **Ecological Principles**
 - Introduction to agro-ecology and biodiversity
 - Principles of agro-ecology and biodiversity
 - The core components of agro-ecology
- **Soil Management**
 - Land preparation techniques
 - Soil aeration
 - Types of seed bed preparation
 - Composting and soil nutrition: Techniques for making, managing, and applying compost to build long-term soil health using local materials
- **Pest Control**
 - Natural pest control using beneficial insects
 - Intercropping
 - Organic remedies
- **Agroforestry and Apiculture**
 - Introduction to agroforestry, importance to Kenya's social economy
 - Nursery types, in agroforestry
 - Principles and importance of tree planting in agroforestry
 - Agroforestry practices and systems – hedgerows, alleys/ spacing, cropping, stripping, rotational fallow
 - Selection and management of tree species in agroforestry

- Bee-keeping: Apiculture: Behaviour, biology and nutrition of honey bees; Beehive types; traditional and modern; setting an apiary; Beekeeping equipment; Harvesting, processing and utilization of products; Pests and diseases of bees and their control.
- **Food Sovereignty: Family Nutrition and Health**
 - Introduction to the principles of human nutrition and world food problem with emphasis on East Africa
 - Nutrients and their sources: Energy requirements and metabolism; Vitamins and minerals; Functions of nutrients in the human body; proteins; Carbohydrates; Fats.
 - Nutrients in the balanced diet.
 - Malnutrition in children: symptoms, causes and cure.
 - Different assessment (Methods) of nutritional status.
 - Foods, preparation, preservation, utilization and storage.
 - Introduction to the use of food consumption surveys, tables and nutritional status survey
 - Family life education.
- **Market**
 - Economics and Marketing: Record keeping, business planning, farm finance,
 - Marketing and value addition of animal products.
 - Use of farm management software and computer applications.

Key Learning Outcomes

- An integrated/holistic understanding of food systems based on agro-ecology/agro-biodiversity principles and practices to contribute to agro-ecological transformation.
- A community of practice established assuring a long –term implementation of agro-ecology.
- Catalytic relations and powerful initiatives developed through well coordinated networks of alumni

Typical Course curriculum

- **Duration:** At one week, two weeks, one month, three months, nine months, one year modular courses
- **Assessment:** Mix of lectures, laboratory experiments, field studies, participatory on-farm demonstrations, on-farm research projects, and examinations. Field work involves practical and experiential learning of different agro-ecological principles and practices. This course can be carried out through e-learning platform, face-to-face sessions, and field work
- **Requirements:** Background in biology, chemistry, mathematics, or geography.

4. Permaculture Course

A permaculture designed course is a standardized curriculum focused on ecological design, sustainability, and regenerative practices. Key modules include ethics (earth care, people care), design methodologies (zones, sectors), water management, soil health, and food forests. Course often ends with a practical design project.

Core Curriculum

- **Introduction and Ethics**
 - History of permaculture
 - Key problems, and
 - The core ethics: Care for earth, people, and fair share/surplus.
- **Permaculture Principles**
 - Exploration of Holmgren's and Mollison's principles of sustainable living.
- **Design Methodologies**
 - Approaches to mapping
 - Analysis (sectors, zones, elevation), and
 - Observing natural patterns
- **Water Management**
 - Harvesting, storing and managing water, understanding the water cycle
- **Soil Health and fertility**
 - Soil structure
 - Composting, and
 - Improving soil fertility
- **Food Forests and Gardening**
 - Designing perennial food systems
 - Companion planting, and
 - Seed saving
- **Ecological Structures**
 - Building techniques and renewable energy systems
- **Social Permaculture**
 - Community development
 - Local economies, and
 - Governance
- **Practical Design Project**

- Students apply learning to create a design project, often for a real –world client.

Structure

- **Duration:** Typically a 72-hour curriculum
- **Delivery:** Can be taught over two weeks, as a series of weekends, or online
- **Focus:** Practical application of ecological principles to homes, landscapes, and communities.

5. Sustainable Agriculture Course

A comprehensive sustainable agriculture course typically covers ecological farming methods, soil health, soil and water conservation, ecological pest management, and economic viability. Key modules include organic farming, Integrated Pest management (IPM), regenerative farming techniques, farm business planning, crop diversification to improve biodiversity management and resilience, and integrated livestock systems, aimed at creating environmentally sound and profitable food production systems. Courses focus on long-term food production with minimal environmental impact.

Course curriculum Modules

- **Module 1: introduction to sustainable Agriculture**
 - Principles of sustainability
 - Whole farm planning and land management
 - Organic farming vs. Conventional systems
 - Natural farming techniques and minimum tillage
- **Module 2: Soil conservation and health**
 - Soil structural improvement and adding organic matter
 - Conservation tillage and cover crops
 - Controlling erosion with stones/soil bunds
- **Module 3: Water Management and conservation**
 - Irrigation systems (drip, precision agriculture)
 - Water –saving measures and recycling
 - Swales and keyline water harvesting
- **Module 4: Agro-ecology and Biodiversity**
 - Integrated pest Management
 - Biodiversity enhancement (wildlife corridors, windbreaks)
 - Weed Management

- **Module 5: Economic and Social Sustainability/viability**

- Farm business planning and marketing
- Economic Principles and value addition
- Social aspects: Sustainable development goals(SDG 2.4.1)

- **Module 6: Regenerative farming techniques**

Regenerative farming practices are farming methods aimed at restoring soil health, increasing biodiversity, and sequestering carbon to combat climate change. Key approaches minimizing soil disturbance (no-till, using cover crops, diversifying crop rotations, integrating livestock, and applying organic fertilizers. These methods build top soil, enhance water retention, and reduce dependence on synthetic inputs.

Key regenerative practices

- **Minimized Mechanical tillage:** Avoiding plowing or digging to protect the soil structure, fungal networks, and microbial life, also known as “no-till” farming.
- **Cover Crops:** Planting cover crops, green manure, or maintaining crop residues during fallow periods to keep the ground covered and prevent erosion.
- **Biodiversity Enhancement:** Utilizing complex crop rotations, intercropping, and creating pollinator strips or hedge rows to increase biodiversity.
- **Livestock Integration:** Integrating livestock into arable farming systems to naturally fertilize the land and help build healthy, carbon – rich soil.
- **Organic Soil Additions:** Applying compost and solid manure to improve soil nutrient levels, structure, and water retention.
- **Living Roots Year Round:** Ensuring that plants are growing in the soil consistently to fuel soil biology and improve soil armour.

Benefits of Regenerative agriculture

- **Climate Change Mitigation:** Increased carbon sequestration in the soil
- **Improved Productivity:** Increased nutrient density in food and higher resistance to drought and erosion.
- **Cost Reduction:** Decreased reliance on chemical fertilizers and pesticides
- **Resilient Ecosystems:** Improved water infiltration **and enhanced habitat for beneficial wild life.**
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- **Module 7: Integrated livestock systems**

- **Introduction to animal production:** Covers the importance of livestock, animal behaviour, and different production systems (Intensive vs. Extensive). Overview of livestock types, importance of livestock industry, and differences between indigenous and exotic breeds.
- **Animal Nutrition and Feeding:** Studies digestive systems, nutrient requirements, feed types (forages, concentrates), and ration formulation for different livestock species.

- **Breeding and Genetics:** Introduction to Mendelian laws, selection methods, artificial insemination, and improving economic traits.
- **Livestock Management Practices:** Routine duties including hoof trimming, docking, dehorning, shearing, identification methods, and milking procedures.
- **Animal Health and Disease Control:** Preventive measures, vaccines, parasite control (dipping/spraying), and common diseases.
- **Housing and Environment:** Design and management of animal shelters, ventilation, hygiene, and environmental impact on production.
- **Production systems:** Analysis of intensive vs. Extensive systems, and organic animal production.
- **Economics and Marketing:** Record keeping, business planning, farm finance, and marketing products.

Key Competencies

Upon completion, learners typically understand how to reduce reliance on agro-chemicals, improve land management practices, and establish economically sound, eco-friendly farms

Structure of Sustainable Agriculture Course

- **Introductory:** 5-day workshops covering basics
- **Comprehensive:** 9-week or 9-Saturday intensive, hands –on courses
- **Methods:** Classroom lessons paired with on –farm, participatory demonstrations

6. Grow Bio intensive Course

A bio-intensive agriculture course focuses on sustainable, small scale farming techniques to maximize yield and soil fertility without chemical inputs, usually covering double-digging, composting, intensive spacing, and carbon/calorie farming. Training typically includes both classroom theory and practical, hands on demonstrations.

Core Module Outline

- **Introduction to Bio-intensive Farming**
 - Principles of sustainable agriculture
 - History, and
 - The 8 core components (Composting, double digging, intensive planting, companion planting, carbon farming, calorie farming, open-pollinated seeds, and whole-system design).
- **Soil Fertility and Preparation**
 - Double digging technique to break hardpans
 - Soil aeration, and

- Creation of raised beds
- **Composting and Soil Nutrition**
 - Techniques for making, managing, and applying compost to build long-term soil health using local materials.
- **Intensive Planting and Spacing**
 - Planting in raised beds using hexagonal spacing to maximize yield per square meter,
 - Conserve water, and
 - Reduce weeds
- **Carbon and Calorie Farming**
 - Dedicated planting for soil sustainability (60% for carbon/compost) and human nutrition (30% for calorie-dense crops)
- **Seed Production and Management**
 - Saving open-pollinated seeds and selecting for vigour, yield, and adaptability
- **Companion Planting and Pest Management**
 - Natural pest control using beneficial insects
 - Intercropping, and
 - Organic remedies
- **Farm Planning and Management**
 - Developing planting calendars
 - Maintaining records, and
 - Scaling for family nutrition and income

Typical Training programs

- **Introductory:** 5-day workshops covering basics
- **Comprehensive:** 9-week or 9-Saturday intensive, hands –on courses
- **Methods:** Classroom lessons paired with on –farm, participatory demonstrations

7. Horticulture

A horticulture course covers plant propagation, cultivation techniques, garden design, and post-harvest management, designed to equip students with skills in fruit, vegetable, and ornamental plant production. Typical curricula include botany basics, greenhouse management, and hands-on training for roles in agricultural research, management, or design.

Core Horticulture Course Modules

- **Plant Science Fundamentals**
 - Botany basics
 - Plant anatomy
 - Soil health
 - Plant nutrition, and
 - Growth physiology
- **Plant Propagation**
 - Techniques for sexual (seeds) and asexual (grafting, cuttings, layering) plant propagation.
- **Crop Management**
 - Commercial production of fruits (Pomology)
 - Vegetables (Olericulture), and
 - Flowers (Floriculture)
- **Plant Protection**
 - Identification and management of plant pests, diseases, and weeds
- **Landscape Technology**
 - Principles of landscaping
 - Site analysis, and
 - Maintenance
- **Post-Harvest Technology**
 - Handling
 - Storage
 - Marketing of horticultural products to reduce losses

Common Course Structure

- **Theory and Practice:** Modules are typically split between classroom and laboratory/greenhouse work
- **Industrial Attachment:** Hands-on experience through field work in horticultural enterprises
- **Electives:** Options may include biotechnology, medicinal plant production, and agroforestry.

Career Outcomes

Graduates often work as horticultural technicians, managers for greenhouses/plantations, landscapers, or in agricultural consultancy.

8. Food Processing Technology

This course provides comprehensive knowledge of food science, preservation, safety, and industrial processes, covering topics from raw material preparation to packaging. Key areas include thermal processing, fermentation, quality control, and sustainable technologies to ensure food safety and quality for industrial applications.

Core Modules and Topics

- **Module 1: Introduction to Food Science & Technology**
 - Scope and trends in the food industry
 - Basic food constituents: carbohydrates, proteins, fats, water, vitamins, and minerals.
- **Module 2: Raw Material Handling & Preparation**
 - Cleaning, sorting, grading, and peeling techniques.
- **Module 3: Food Preservation Techniques**
 - Low-temperature preservation (cooling, Refrigeration, Freezing)
 - Thermal processing (Pasteurization, Sterilization, Blanching, UHT).
 - Drying and dehydration technologies
 - Fermentation and chemical preservatives.
- **Module 4: Specialized Food Processing Technologies**
 - Cereal processing (milling, baking)
 - Fruits and vegetable processing
 - Animal products processing (meat, dairy, fish)
 - Fats and oils technology
- **Module 5: Food safety, Quality Control, and Management**
 - HACCP (Hazard Analysis and Critical Control Points) and food safety systems
 - Quality assurance, hygiene, and sanitation in processing units
 - Sensory evaluation and food analysis techniques
- **Module 6: Packaging and storage**
 - Packaging materials and technologies
 - Shelf-life testing and storage regulations
- **Module 7: Advanced and Emerging Technologies**
 - High-pressure processing, Ohmic heating, pulse electric field processing.
 - Sustainable food processing

Practical & Project Component

- **Laboratory sessions:** Hands-on experience with unit operations, testing, and processing equipment.
- **Industrial Training/Internships:** In-plant training for practical skills in industrial settings
- **Product Development:** Project work on creating new food products.

Key Learning Outcomes

- Understand and apply principles of food chemistry, microbiology, and engineering
- Operate food processing machinery and apply preservation techniques
- Implement safety and quality standards (HACCP)
- Manage food processing waste and ensure sustainability.

Typical Training Format

- **Introductory:** 5-day workshops covering basics
- **Comprehensive:** Module based training
- **Methods:** Classroom lessons paired with participatory demonstrations/practical sessions

9. Community Development

A community development course equips practitioners with skills to plan, implement, and evaluate community initiatives, focusing on empowerment, social justice, and sustainable, participatory approaches. Key topics include needs assessment, project management, resource mobilization, and leadership, preparing students and participants for roles in NGOs, government agencies, and community based organizations.

Core Module Topics

- **Introduction to Community Development**
 - Concepts, principles, and history of community empowerment and change
- **Community Needs Assessment and Mapping**
 - Identification of community needs, situational analysis, and participatory rural appraisal techniques.
- **Project Planning and Management**
 - Design, implementation, and management of development projects, including project life cycle, Monitoring and Evaluation (M&E)
- **Participatory Approaches**
 - Strategies for ensuring community engagement, ownership, and stakeholder mobilization.

- **Resource Mobilization and Fundraising**
 - Techniques for securing resources for community initiatives, including grant writing and community fundraising
- **Leadership and Conflict Management**
 - Developing leadership skills, community group formation, and managing conflict within groups.
- **Cross-cutting Issues**
 - Inclusion of gender equality, human rights, environmental sustainability, and health education in development

Course Structure Highlights

- **Foundation Units:** Sociology, Economics, and African Social Structures.
- **Skills Units:** Report writing, advocacy, communication skills, and data analysis
- **Specialized Areas:** Disaster management, microfinance, urban/Rural development, and Policy Advocacy

Targeted Skills

Graduates gain proficiency in social research, community mobilization, advocacy, and strategic planning, making those effective leaders in enhancing community welfare.

2. Hospitality and institutional management

I. Baking Technology

Baking Technology curriculum, often structured as a diploma or certificate (e.g. TVET CDACC level 4-5), provides comprehensive training in commercial bread making, pastry production, and confectionery. Key areas include ingredient science, dough management, specialized techniques (laminated dough), cake decorating, food safety, and business management.

Core curriculum components

- **Yeast- Leavened Products:** Techniques for white bread, sweet bread, speciality breads, and dough management.
- **Pastry & Confectionery Production:** Puff pastry, short crust, choux, tarts, cookies, and laminated dough.
- **Cake Making & Decoration:** Specialized training in cake types, icing, piping, fondant, and multi-tiered wedding cake assembly.
- **Baking Science & Technology:** Understanding ingredient functionality (Flour, yeast, fats) and chemical reactions.
- **Commercial Operations & Safety:** Food safety standards (HACCP), bakery machinery maintenance, costing, and recipe structuring.

Course Structure & Focus:

- **Practical Focus:** High emphasis on hands-on training (often 60-80% practical), commercial workshop training, and industrial attachments.
- **Modes of Study:** Typically 3 months to 2 years, depending on whether it is a proficiency certificate or diploma level.
- **Career Outcomes:** Graduates are qualified as pastry chefs, commercial bakers, bakery managers, or entrepreneurs.

Common Curriculum Subjects:

- Food Safety and Hygiene

- Baking Equipment Operations
- Industrial Baking Techniques
- Recipe Development and Costing
- Traditional and International Pastry Techniques

The curriculum frequently integrates modern baking trends and specialized dietary requirements, such as gluten-free or healthier baking alternatives.

If you are looking to enrol:

- Compare certificate versus diploma options
- Understand the entry requirements (e.g. KCSE Grade C- or similar)

II. Accommodation operations

An Accommodation Operations Course curriculum focuses on the management, maintenance, and service delivery of lodging facilities, combining housekeeping, front office operations, and guest services. Core topics include room cleaning, reservation systems, property management systems (PMS), security, and facility management, usually featuring practical industry management.

Core Curriculum Components:

- **Housing keeping Management:** Room inspection, cleaning techniques, linen management, laundry operations, cleaning equipment and agents, and interior decoration.
- **Front Office Operations:** Guest checking/check-out procedures, reservations, handling guest inquiries, concierge services, and night auditing.
- **Hospitality Technology:** Use of computerized Property Management Systems (PMS) for reservations, room assignment, and billing.
- **Security and Safety:** Key control protocols, emergency procedures, health and safety regulations, and security of guest property.
- **Interpersonal and Management Skills:** Professional etiquette, customer service excellence, stress management, staff scheduling, and inventory control.
- **Practical Training/Industrial Attachment:** Typically includes a 320-hour (or longer) internship in a hotel or hospitality establishment to gain real-world experience.

Common Course Levels:

- **Certificate:** Focuses on operational roles like room stewards and front desk attendants

- **Diploma:** Covers supervisory and management skills in addition to operations.

Important:

If you'd like, I can:

- Find the full, detailed syllabus for a specific certification level (Like level 4/5)
- Highlight specific modules on sustainability on sustainability or revenue management.

III. Cooking Skills

A comprehensive cooking skills curriculum covers fundamental knife handling, cooking methods (roasting, sautéing, boiling), Kitchen safety/hygiene, and flavour development. Programs often include **mise en place**, **menu planning**, **stock/soup preparation**, and **nutrition**. These courses cater to beginners up to professional levels, teaching students to cook from scratch and understand food components.

Curriculum Breakdown:

Foundation of Cooking Course Curriculum

- **Module 1: Kitchen Basics**
 - **Safety & Hygiene:** Sanitation, food handling, HACCP, and allergen awareness.
 - **Equipment Identification:** Understanding commercial/home kitchen equipment.
 - **Kitchen Organization:** Mise en place (setup), workflow, and efficiency.
- **Module 2: Knife Skills and Prep**
 - **Knife Handling:** Safe handling, maintenance, and sharpening
 - **Cutting Techniques:** Proper dicing, slicing, julienne, and batonnet techniques
 - **Vegetable Prep:** Cleaning, prepping, and storing produce.
- **Module 3: Cooking Methods and Techniques**
 - **Dry Heat Methods:** Roasting, baking, grilling, searing, and sautéing.
 - **Moist Heat Methods:** Boiling, simmering, poaching, and steaming.

- **Fat-Based Methods:** Shallow frying, deep frying, and sweating.
- **Module 4: Stocks, Soups, and Sauces**
 - **Stock Production:** Making chicken, vegetable, and beef stocks.
 - **Soups:** Cream soups, broths, and consommés.
 - **Mother Sauces:** Béchamel, Veloute, Espagnole, Tomato, and Hollandaise.
- **Module 5: Ingredients and Flavor Development**
 - **Flavor Profiles:** Using herbs, spices, and aromatics.
 - **Salt, Fat, Acid, and Heat:** Understanding the essential components of flavor.
 - **Menu Planning:** Balancing texture, color, and taste in a meal.
- **Module 6: Specialized Components**
 - **Starch/Grains:** Rice, pasta, potato cookery
 - **Protein Cooking:** Poultry, meat, and fish preparation techniques.
 - **Salads/Dressings:** Cold preparations and emulsification.
- **Module 7: Introduction to Pastry and Baking**
 - Basic pastry doughs, breads, or simple desserts
- **Module 8: Final Assessment**
 - Practical demonstration of skills, such as preparing a three-course meal

Key Skills Developed:

- **Confidence** in the Kitchen
- Ability to **read and adapt recipes**
- **Menu planning** and basic costing

Important: Would you like to narrow this down to a specific type of course, such as **home cooking, culinary arts, or a specialized baking/pastry** class?

IV. Cosmetology

A comprehensive cosmetology course curriculum prepares students for licensure by covering hair design, chemical services, skincare, nail care, and salon management, typically involving 1,000 - 1600 + hours of training. Key areas include **sanitation** (sterilization techniques), haircutting/styling, colouring, chemical reformation (perms/relaxers), **Makeup application**, and **spa treatments**.

Key Curriculum Modules and Components

- **Hair Care & Styling:**

Shampooing/conditioning, plaiting, braiding, thermal styling, wig application, and hair extensions.

- **Chemical Services:** Permanent waving, chemical hair relaxing, and hair coloring techniques.

- **Skincare & Beauty:** Facial treatments, skin analysis, eyebrow shaping, eyelash treatment, and hair removal (waxing).

- **Nail Care:** Manicure, pedicure, and nail art.

- **Professional Practices:** Salon health and safety procedures, salon management, and customer service skills.

Training Structure

- **Theory & Hands-on:** Instruction includes lectures, demonstrations, and practical, hands-on practice, often using mannequins before working on clients in a clinic setting.

- **Safety Protocols:** Intensive focus on sterilization, sanitation, and infection control in compliance with state regulations.

- **Professional Development:** Training on career planning, communication, and professional ethics

Programs are typically structured as Grade 3 (12 months up to Grade 1 (6 Months) for advanced training, often aligned with national vocational standards.

Can inquire on:

- Detail the specific hairdressing or nail technology modules
- Provide a breakdown of typical hourly requirements for state boards.

V. Fashion Designer

A fashion course design course covers the end-to-end process of creating clothing and accessories, combining creative design with technical skill. Key components include fashion illustration, pattern drafting, garment construction, textile science, trend forecasting, and portfolio development, often incorporating digital design tools. Programs range from 6-months upwards.

Core Curriculum Components:

- **Design & Illustration:** Fashion sketching, figure drawing, color theory, mood boards, and digital design using software.
- **Garment Construction & Technical Skills:** Pattern making, cutting, sewing techniques, garment finishing, and fitting for men's, women's, and children's wear.
- **Textile Science:** Study of fabrics, properties (durability, texture), and fabric decoration techniques.
- **Fashion Theory & Marketing:** Trend analysis, forecasting, brand development, and fashion merchandising.
- **Industry Practice:** Portfolio development, studio management, and entrepreneurship.

Typical Course Structure:

- **Certificates (6 – 12 months):** Focus on basic tailoring, fashion crafts, and hands-on sewing
- **Diploma (2-3 years):** Cover advanced design techniques, pattern drafting, and merchandising.

Career Pathways:

Graduates can pursue careers as fashion designers, fashion illustrators, pattern makers, stylists, textile designers, or entrepreneurs.

VI. Food and Beverage operations

A Food and Beverage (F & B) course curriculum, such as those offered by KNEC or specialized hospitality schools, covers both production and service management. Key areas include culinary techniques, menu planning, cost control, beverage management, and health & safety standards. It typically combines practical hands-on training with business management principles to prepare students for roles in hotels, restaurants, and catering sectors.

Core Curriculum Components:

- **Food and Beverage Production (Culinary Arts):** Fundamentals of cooking techniques, recipe development, international cuisine, and food safety/hygiene.
- **Food and beverage Service:** Restaurant service techniques, banquet management, menu design, and customer service excellence.
- **Management & Operations:** Food & beverage control systems, inventory management, menu engineering, and purchasing.
- **Financial & Business Skills:** Hospitality accounting, cost control calculations, budget planning, and marketing strategies.
- **Hospitality Management:** Human resource management, hospitality law, communication skills, and health & safety practices.

Common Course Levels:

- **Certificate:** Focuses on operational skills, service techniques, and basic safety.
- **Diploma:** Focuses on advanced managerial-level skills, including cost control and strategic operations.

Practical Training

The curriculums emphasize hands-on training, supervised restaurant or kitchen management, and professional attachments.

Inquiries can be made on:

- Whether you are looking for a certificate or a diploma level?
- Are you interested in the production (cooking) side or the service/management side?
- Do you need it for an examining body (like KNEC)?

3. Tailoring Course Curricula

Key Components of Tailoring Courses:

- **Fundamental skills:** Introduction to sewing tools, machine operation, safety regulations, and maintenance.
- **Techniques:** Hand and machine stitching, sewing seams, hems, and finishing techniques.
- **Design and Drafting:** Pattern drafting for various styles, garment pattern modification, and understanding fashion design principles.
- **Material Selection:** Understanding fabric properties, handling materials, and fabric cutting
- **Garment Construction:** Creating garments for men and women, including shirts, trousers, skirts, and dresses.
- **Advanced Skills and Business:** Advanced alteration, garment repair, tailoring entrepreneurship, fashion portfolio development, and costing for business.

Common Course Structure Examples:

- **Short Course (Beginners):** 6 – week module covering basic stitches, machine operation, patterns, fabric, and creating a simple garment like a skirt.
- **6 – Month Program:** Covers sewing machine operation, pattern making, and stitching various garments.
- **Vocational Training (Modular):** Often split into Grades III, II and I, increasing in complexity from basic stitching to advanced garment construction and costing.

4. Engineering/Agricultural engineering

I. Building Technology

Building technology courses focus on practical construction skills, covering masonry, carpentry, technical drawing, structural analysis, site management, and quantity surveying over 1.5 – 3 years. Curricula are regulated by KNEC or TVET CDACC, featuring modular units and industrial attachment, typically requiring KCSE or D + for entry.

Key Curriculum Areas

- **Core Construction:** Masonry, carpentry, floor tiling, plumbing, and structural concrete technology.
- **Technical Drawing:** Interpretation and creation of construction blueprints, CAD (Computer- AIDED Design).
- **Management & Safety:** Site management, project scheduling, estimation, cost control, and OH & S (Occupational Health & Safety).
- **Theory & Science:** Construction materials, applied science, mathematics, and surveying.

Typical Program Structure

- **Craft Certificate (KNEC):** Generally a 1.5 – 2 year program designed for site supervision and artisan skills, featuring modules on construction materials, site management, and structural basics.
- **Diploma in Building Technology (TVET):** A 3-year Level 6 program covering advanced topics like construction management, services (mechanical/electrical), and computer – aided design.

Common Core Units

1. **Building Construction I & II**
2. **Construction Materials** (Science & Technology)
3. **Technical Drawing/CAD**
4. **Estimating and Costing**
5. **Site Management & Supervision**
6. **Structural Mechanics**
7. **Entrepreneurship & Communication Skills**

II. Carpentry and Joinery

A Carpentry and Joinery course curriculum focuses on building practical woodworking skills, safety, and technical knowledge for construction and furniture manufacturing. Core modules include using hand/power tools, reading technical drawings, timber technology, site carpentry (roofing/framing), and shop joinery (doors/cabinets). Programs range from introductory certifications to advanced diplomas.

Core Curriculum Components

- **Workshop Safety & Tool Maintenance:** Proper use of hand tools (chisels, saws) and portable power tools (routers, circular saws).
- **Timber Technology:** Understanding wood types, properties, seasoning, and defects.
- **Joinery Techniques:** Creating traditional joints (mortise and tenon, dovetails) for furniture and fixtures.
- **Structural Carpentry (First Fix):** Constructing timber frames, floor joists, stud walls, and roof trusses
- **Finishing Carpentry (Second Fix):** Installing door frames, window frames, skirting boards, and cabinetry.
- **Reading Plans & Measurement:** Interpreting construction drawing and applying accurate marking-out methods.
- **Health and Safety:** Regulations regarding construction site safety, PPE, and workshop health and safety.
- **Industrial training/Internship:** Practical hands-on experience in a workshop or on a construction site.

Typical certification structure

- **Level 1 & 2 certificates:** Basic skills in furniture making, bench joinery, and site work.
- **Level 3 Diploma/Advanced certificate:** Complex joinery, roof framing, and supervisory skills

Common career paths include becoming a construction carpenter, finish carpenter, cabinet maker, or site supervisor.

Details on level 1/2 and Level 3 curricula can be provided. Also details on apprenticeship vs. Classroom training.

In case you need more information on this contact the centre management.

III. Plumbing

A plumbing course entails comprehensive training in installing, repairing, and maintaining water supply, drainage, and gas systems in residential and commercial buildings. Programs

combine classroom theory with hands – on workshop training on pipefitting, fixture installation, and safety protocols. Key topics include:

- **Pipefitting and joining:** Cutting, bending, and joining materials like PVC, copper, and steel.
- **Systems Installation:** Setting up water supply pipes, drainage networks, sewage systems, and water heating systems.
- **Fixture Installation:** Installing toilets, sinks, faucets, showers, and water tanks.
- **Maintenance and repair:** Diagnosing leaks, clearing blockages, and servicing fixtures.
- **Technical skills:** Reading blue prints, site management, and safety protocols (health and safety standards).

Most courses provide hands on experience, often preparing learners for certification examinations (e.g. KNEC, NITA) in 9 months to 2 years.

IV. Welding and fabrication

A welding and Fabrication course combines classroom theory with hands-on training in metal joining, cutting, and shaping, covering key techniques like SMAW (stick), GMAW (MIG), and GTAW (TIG). Programs typically include blue print reading, safety protocols, material science, and structural fabrication, preparing students for roles in manufacturing, construction, and repair.

Core Curriculum Components

- **Welding Processes:** In-depth training on Manual Metal Arc Welding (MMAW/Stick), Metal Active Gas (MAG/MIG), Tungsten Inert Gas (TIG), and Oxy-acetylene welding.
- **Fabrication Techniques:** Techniques for cutting, bending, forming, and assembling metal components, including structural steelwork and sheet metal work.
- **Technical Drawing and Blueprints:** Reading and interpreting engineering drawings, welding symbols, and specifications.
- **Workshop Technology & Safety:** Training on machinery maintenance, proper use of Personal Protective Equipment (PPE), and safety regulations.

- **Material Science:** Understanding the properties of metals (aluminium, stainless steel, mild steel) and their reaction to heat.
- **Quality Control & Inspection:** Identifying weld defects (visual inspection) and ensuring structural integrity.

Module Breakdown

- 1. Introduction to welding & Safety:** Fundamentals, safety regulations, and equipment maintenance.
- 2. General metalwork & Fabrication:** Measuring, marking, cutting, bending, and shaping metal.
- 3. Advanced Welding Techniques:** Specialized, high precision welding on diverse materials.
- 4. Structural Fabrication & Pipe Welding:** Project –based training on creating frameworks, tanks, or structural components.
- 5. Engineering Mathematics & science:** Applied calculations for welding projects.

Certification and Diploma Levels

- **Certificate/NITA Level:** Focuses on foundational practical skills, often 6-12 months
- **Diploma Level:** Includes advanced welding, computer – aided design (CAD), and industrial management, typically lasting 2-3 years

Important Note: are you looking for a certificate program, a diploma program, or short NITA-certified courses? Knowing if you are interested in a specific, shorter course or a longer, more in-depth program will help me narrow down the options.

V. Electrical installation

An electrical installation course provides comprehensive training in residential, commercial, and industrial wiring, emphasizing safety, practical skills, and regulatory compliance (e.g. IEEE/local codes). Core modules include electrical principles, blueprint reading, conduit bending, circuit testing, maintenance, and solar PV technology, typically spanning from 4-months certificate programs to 2-year diplomas.

Core Curriculum Components

The curriculum is designed to produce competent, safe technicians through theoretical and practical training:

- **Electrical Fundamentals:** Basic electrical principles (voltage, current, resistance), Ohm's Law, power calculations, and basic electronics.
- **Safety and Regulations:** Occupational Health & Safety (OHS), grounding, and strict adherence to electrical codes (e.g. PEC standards)
- **Wiring and Installation:**
 - **Domestic & Commercial:** Installation of lighting, switches, sockets, consumer units, and distribution boards.
 - **Industrial:** Motor controls, 3-phase systems, transformers, and specialized machinery.
 - **Conduit and Cabling:** Conduit bending (PVC/Metallic), conduit installation, cable pulling, and cable tray installation
- **Technical drawing and Layout:** Reading, interpreting, and drawing electrical schematics, wiring diagrams, and building plans.
- **Testing and Troubleshooting:** Inspection, testing of insulation resistance, earth fault loop impedance, and fault diagnosis.
- **Specialized Areas:** Solar photovoltaic (PV) installation, HVAC systems, and environmental safety in damp areas.

Typical Course Structure

- **Artisan Level (Level 3):** Basic wiring and safety (Approximately 1 term/3-4 months)
- **Certificate Level (Level 4/5):** Detailed installation, planning, and design (approx 2-4 terms/1 year).
- **Diploma Level (Level 6):** Advanced design, management, and complex systems (up to 2 years)

Practical Skills Assessment

Practical competency is a core requirement involvement:

- Real-world wiring projects in workshops.
- Continuous assessment of wiring quality
- Performance of specific tasks such as installing conduit and pulling cables.
- Final installation projects.

Important: If needed to narrow this down, details can be provided on:

- Specific topics within industrial electrical training.
- National certification in Kenya
- Common careers and salaries after completing course.

VI. Agricultural Engineering

Agricultural engineering courses focus on applying engineering principles to food and fiber production, covering soil/water conservation, farm machinery, irrigation, and post-harvest technology. The curriculum typically spans 4-5 years, combining engineering mechanics with agricultural science and practical industrial attachments to prepare students for mechanization and management roles.

Core Curriculum Components

- **Farm Power and Machinery:** Principles of tractors, internal combustion engines, and agricultural machinery design.
- **Soil and Water Engineering:** Irrigation design, drainage systems, soil mechanics, and hydrology
- **Agricultural Processing & Food Engineering:** Post-harvest technologies, crop drying, storage, and processing.
- **Agricultural Structures:** Design of greenhouses, farm buildings, and rural infrastructure.
- **Engineering sciences:** Thermodynamics, Fluid Mechanics, Engineering drawing, and Engineering Mathematics.
- **Environmental & Bio-systems:** Waste management, renewable energy, and environmental impact assessment.

Typical program Structure

Duration: Certificate course range from short courses to one year

Practical Training: Includes a compulsory external industrial attachment, typically lasting eight weeks.

Specialization: Studies in precision agriculture, machine design, or post-harvest technology

Important: For further details contact the college management

Short Courses

1. Soap Making

- **Safety and Basics:** understanding the science of saponification, handling lye safely, and essential equipment
- **Soap making techniques:**
 - **Cold press:** Mixing oils and lye for a slow- cured, custom soap
 - **Hot press:** Cooking the soap for faster saponification and rustic look
 - **Melt – and - Pour:** Designing sops using pre-made bases
 - **Liquid soap making:** Creating liquid and cream soaps
- **Ingredients and Formulation:** Choosing oils, fats and butters, and learning to calculate recipes
- **Customization:** Incorporating natural colorants, herbs, botanical additives, and essential oils for scent.
- **Design and Troubleshooting:** Learning swirl techniques and fixing common issues like seized soap or false trace.
- **Business Basics:** Labelling, packaging, and local regulations for selling.

2. Agricultural Value Chain Training Skills

- **Introduction to Value Chains:** Definition of value chain actors and stages (input supply, production, processing, and distribution).

- **Value Chain Mapping and Analysis:** Methodologies for mapping chains, identifying nodes, assessing competitiveness, and diagnosing constraints.
- **Market Linkages and Analysis:** Strategies for connecting farmers to markets, understanding demand, and establishing contract farming.
- **Value Addition and Upgrading:** Techniques for product quality improvement, processing, and developing certification standards.
- **Value Chain Governance and Finance:** Understanding power dynamics within Chains, managing relationships, and accessing financial services.
- **Sustainability and Gender:** Integrating social inclusion, gender mainstreaming, and environmental sustainability in value chains.

Key Focus Areas

- **Development of Inclusive Business Models:** Ensuring smallholder farmers benefit from market participation
- **Risk Management:** Identifying and overcoming challenges such as logistics gaps, climate risks, and market fluctuations.
- **Practical Tools:** Utilizing analytical techniques for mapping and assessing chain performance.

N/B: These courses are intended for professionals in agricultural development, agribusiness managers, and policy makers looking to improve market competitiveness and sustainability

3. Computer and Information technology - Computer/ICT/AI Skills

- Computer science
- Cyber security
- ICT technician
- AI

4. Liberal studies

- Counselling Psychology

- Library and Information
- Project management
- Social work and Community Development

5. **Business studies**

- Accountancy
- Banking and finance
- Business management
- Human Resource Management
- Cooperative management
- Marketing management
- Store keeping management
- Supply chain management

Important: Most of the Courses at OTEPIC are done at one week, three weeks, one months, three months, six months and one-year modular programs tailored on the client's needs.

The Course prospectus highlights contents of each area, grades for entry and durations of each course and commencement for each program, which can be obtained upon request.

For Further details contact:

The Director,
JUME OTEPIC